

BELLINGSHAUSEN STATION, AntarcticaWMO: **890500**Lat: **62.192S**Lon: **58.936W**Elev: **46**StdP: **14.67**Time Zone: **-3.00 (W03)**Period: **94-19**WBAN: **99999****Annual Heating, Humidification, and Ventilation Design Conditions**

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	-1.1	3.5	-5.5	4.1	-0.6	-1.0	5.2	3.9	85.4	9.9	75.6	11.5	23.7	160	1.132

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	3.5	39.6	38.1	38.6	37.3	37.9	36.7	38.7	39.2	37.8	38.3	37.1	37.6	18.9	0

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
37.9	33.5	38.6	37.1	32.5	38.0	36.4	31.6	37.4	14.8	39.5	14.3	38.5	14.1	37.7	43.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 59.0	48.9	40.8	DB	-2.6	42.4	5.9	2.3	-6.9	44.0	-10.3	45.3	-13.7	46.6	-18.0	48.3	(4)
(5)			WB	-3.4	40.5	6.1	1.4	-7.8	41.5	-11.4	42.3	-14.8	43.1	-19.2	44.1	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	28.1	34.3	34.7	33.1	28.9	27.2	23.3	21.1	21.4	23.8	27.5	29.7	32.3
	DBStd	7.12	2.04	2.59	3.38	5.44	5.13	7.02	8.41	8.07	6.89	4.14	2.99	2.23
	HDD50	7999	485	429	523	632	705	801	897	887	787	697	608	548
	HDD65	13474	950	849	988	1082	1170	1251	1362	1352	1237	1162	1058	1013
	CDD50	0	0	0	0	0	0	0	0	0	0	0	0	0
	CDD65	0	0	0	0	0	0	0	0	0	0	0	0	0
	CDH74	0	0	0	0	0	0	0	0	0	0	0	0	0
	CDH80	0	0	0	0	0	0	0	0	0	0	0	0	0
Wind	WSAvg	17.6	15.4	15.9	16.5	18.7	17.7	19.1	18.7	18.7	18.8	19.5	17.1	15.3
Precipitation	PrecAvg	40.9	3.2	3.2	4.7	4.0	3.8	3.0	2.8	3.0	3.2	3.9	3.3	2.8
	PrecMax	54.8	4.5	4.9	7.4	6.0	6.3	4.5	4.3	5.6	5.4	6.7	4.5	4.5
	PrecMin	33.7	1.9	1.9	2.8	2.3	1.7	1.6	1.3	0.6	1.1	2.0	1.2	1.2
	PrecStd	4.6	0.6	0.7	1.1	0.9	1.1	0.7	0.9	1.1	0.9	1.0	0.8	0.8
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	41.1	41.2	41.0	38.4	36.6	35.0	33.8	33.6	34.2	34.9	36.9	39.1
		MCWB	39.4	39.5	39.3	37.0	35.3	33.8	32.7	32.4	32.9	33.5	35.3	37.4
	2%	DB	39.0	39.6	39.1	36.9	35.4	33.8	32.7	32.4	33.0	33.7	35.0	37.3
		MCWB	37.4	38.2	37.7	35.7	34.3	33.0	31.9	31.6	32.2	32.6	33.8	35.7
	5%	DB	38.0	38.7	38.2	35.8	34.4	32.8	31.9	31.7	32.3	32.9	34.1	36.2
		MCWB	36.7	37.5	37.1	34.9	33.6	32.1	31.2	31.0	31.7	32.0	33.1	34.7
	10%	DB	37.3	38.0	37.4	34.9	33.6	31.9	30.9	30.9	31.6	32.2	33.3	35.4
		MCWB	36.1	36.8	36.4	34.0	32.9	31.3	30.3	30.3	31.1	31.5	32.4	34.0
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	39.9	40.1	39.6	37.5	35.7	34.1	33.1	33.0	33.6	34.0	35.7	37.8
		MCDB	40.5	40.8	40.3	38.0	36.1	34.4	33.3	33.2	33.9	34.4	36.5	38.5
	2%	WB	38.1	38.7	38.4	36.0	34.7	33.2	32.2	32.0	32.4	33.0	34.2	36.1
		MCDB	38.6	39.2	39.0	36.5	35.1	33.5	32.5	32.2	32.7	33.3	34.6	36.7
	5%	WB	37.2	37.8	37.5	35.1	33.9	32.3	31.4	31.3	31.9	32.3	33.3	35.2
		MCDB	37.7	38.4	37.9	35.6	34.3	32.6	31.7	31.5	32.1	32.6	33.7	35.8
	10%	WB	36.5	37.2	36.6	34.3	33.1	31.4	30.4	30.4	31.2	31.7	32.7	34.4
		MCDB	37.0	37.8	37.2	34.8	33.5	31.8	30.8	30.8	31.5	32.0	33.2	35.1
Mean Daily Temperature Range	5% DB	MDBR	3.4	3.5	4.1	4.9	5.4	6.5	7.9	7.3	6.6	4.4	3.6	3.2
		MCDBR	4.4	4.2	4.3	4.4	5.0	5.3	6.1	5.7	5.0	3.8	3.8	4.2
		MCWBR	3.9	3.6	4.1	4.4	4.8	5.4	6.1	5.8	5.2	3.8	3.6	3.5
	5% WB	MCDBR	4.1	4.0	4.1	4.1	4.8	5.1	6.0	5.8	4.9	3.6	3.7	3.9
		MCWBR	3.9	3.8	4.2	4.4	4.6	5.2	6.4	6.0	5.1	3.8	3.6	3.6
Clear-Sky Solar Irradiance	taub	0.309	0.308	0.297	0.280	0.225	0.366	0.211	0.269	0.288	0.310	0.304	0.309	
	taud	2.493	2.501	2.501	2.373	2.100	2.821	2.049	2.239	2.369	2.400	2.457	2.465	
	Ebn at Noon	307	291	267	214	141	96	139	209	263	286	306	310	
	Edh at Noon	31	28	23	17	10	6	10	18	25	30	32	33	
All-Sky Solar Radiation	RadAvg	1589	1146	617	270	78	33	56	201	529	981	1503	1800	
	RadStd	146	88	51	22	7	1	3	17	42	65	91	144	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (7 neighbors)	N/A	N/A	N/A	N/A	-0.47	-0.76	N/A	N/A	N/A	N/A

Nomenclature: See separate page